

Design Specification: PLPI Assembly Control Software

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DESIGN SPECIFICATION

PLPI Batch Record Automation

Phase 2: B&S Batch Add, Printing Modules and Leaflet Folding

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1. Introduction

This Design Specification defines the detailed PLPI design for Phase 2 of the Batch Record Automation project. It converts the approved user and functional requirements into screen, control, validation, workflow, data, audit and routing behaviour for B&S Batch Add, BAR generation, Label Printing, Leaflet Printing, Carton Issuing, Braille Printing and Leaflet Folding.

1.1 Purpose of the Document

The document provides the implementation-level design required by business, QA, IT, development, testing and operational stakeholders. It identifies each visible control, whether it is Existing, New or Existing with updated behaviour, and defines the enablement, validation, record creation, audit and downstream handoff associated with the control.

1.2 Scope

The design starts when an eligible source record is available in B&S Batch Add. It covers compatible record selection, one controlled electronic BAR, B&S verification and line clearance, the shared Printer menu and queue design, Label/Leaflet/Carton/Braille processing, the new unified Print pop-up and completion controls, Leaflet Folding and the controlled handoff to Pre-Assembly. The BAR includes downstream placeholders and audit pages, but detailed Pre-Assembly, Production Control, Assembly, Post-Assembly QC, Pre-QP and QP design is outside this phase.

1.3 Reference Documents

Reference	Use
Final URS - PLPI BAR: B&S Batch Add, Printing and Leaflet Folding	Approved business/user requirements and scope.
Final FS - PLPI BAR: B&S Batch Add, Printing and Leaflet Folding	Approved functional behaviour and field classification.
Current project wireframe (wireframe/app.js)	Detailed screen controls, visible values, button states, validation messages, data objects and workflow routing.
DS-PLPI BAR reference/template	Page system, typography, headings, tables, header/footer and presentation pattern.

1.4 Abbreviations

Abbreviation	Definition
B&S	B&S Batch Add / controlled batch identifier
BAR	Batch Assembly Record

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Abbreviation	Definition
DS	Design Specification
ECMA	Approved product/artwork reference
FS	Functional Specification
GMP	Good Manufacturing Practice
IT	Information Technology
MFG	Manufacturing / manufacturer
PCL	Product Check Log
PDF	Portable Document Format
PLPI	Parallel Import / PLPI workflow system
QA	Quality Assurance
QP	Qualified Person
RP / RPi	Responsible Person / Responsible Person Import
URS	User Requirement Specification

2. Overall Design

Phase 2 is implemented inside the existing PLPI application. Existing screens and visible fields are replicated where identified; new controlled behaviour is layered onto them without changing the established application navigation. All actions use the authenticated PLPI session and are retained against the B&S batch and controlled BAR.

2.1 End-to-End Workflow

Sequence	Stage	Design outcome
1	B&S Batch Add	Search eligible source records, select one or compatible matching records and request BAR generation.
2	BAR Generation	Confirm generation, create one populated 13-page BAR, retain source-record links and creation attribution.
3	B&S Line Clearance	Complete three electronic checks, comments where applicable and authenticated User Sign Off.
4	Label Printing	Complete all route-derived label lines and Print Done.
5	Leaflet Printing	Where required, open the controlled leaflet PDF, complete the applicable test/master and actual print, then Print Done.

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Sequence	Stage	Design outcome
6A	Carton Issuing	For Reboxing, confirm required carton issue and any controlled extra cartons.
6B	Braille Printing	For applicable Relabelling, complete the braille label and declaration-copy rows.
7	Leaflet Folding	After all route-required preparation, confirm folding completion and attribution.
8	Pre-Assembly handoff	Remove the completed item from the active folding queue and expose it to the configured downstream stage.

2.2 Route Logic

Route / condition	Required sequence after B&S sign-off	Eligibility rule
Reboxing with leaflet	Label Printing -> Leaflet Printing -> Carton Issuing -> Leaflet Folding	Carton Issuing must be complete before folding.
Relabelling with leaflet and braille	Label Printing -> Leaflet Printing -> Braille Printing -> Leaflet Folding	Braille Printing must be complete before folding.
Relabelling with leaflet and no braille	Label Printing -> Leaflet Printing -> Leaflet Folding	No carton or braille stage is inserted.
No leaflet required	Label Printing -> next configured stage	Leaflet Printing and Leaflet Folding are bypassed according to route configuration.

2.3 Roles and Access

Role	Permitted design actions
B&S Batch Add / BAR Creation User	Search, select, generate/view/print BAR, complete line-clearance checks and sign off.
Printer User	Open Printer menu and eligible queues; access approved documents; perform Label, Leaflet, Carton and Braille activities assigned to the role.
Leaflet Folding User	Search eligible batches, review product/document context and confirm folding completion.
QA / RP / Operations / IT	Controlled review, exception, reporting or support access as authorised; no operational completion unless the assigned role permits it.

2.4 Common Design Principles

Identity: B&S Batch Number is the primary workflow key. MFG Lot No., source record IDs, product identifiers and document references remain linked.

Control classification: Existing means replicated current-system presentation. New means new electronic control/data. Existing - updated means replicated presentation with new validation, audit or routing.

Status model: Active identifies untouched eligible work; In Progress identifies saved or partially completed work; Done/Completed identifies a locked stage; downstream statuses show the current route position.

Attribution: All controlled generation, document print, print run, line completion, Print Done and folding completion events retain authenticated user and date/time.

Data integrity: A completed event is appended to history; later extra prints or corrections do not overwrite the original completion record.

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3. Detailed Design Specification

3.1 B&S Batch Add and BAR Creation

The existing BNS Batch Creation screen is retained as the entry point. The design adds controlled selection, compatibility checking, BAR confirmation, BAR record creation, electronic line clearance and authenticated release to Label Printing.

3.1.1 Search and Result Grid

Control	Classification	Design type	Detailed behaviour
Country	Existing filter	Selection list	Filters by country; blank/All does not restrict the results.
Site	Existing filter	Selection list	Filters by processing site; blank/All does not restrict the results.
Category	Existing filter	Selection list	Values include Reboxing and Relabelling.
Stock	Existing filter	Selection list	Retained current-system stock criterion.
Status	Existing - updated	Selection list	Values: All, Not Printed BAR, Printed BAR.
Batch_No	Existing filter	Text input	Locates a B&S batch by entered value.
Search	Existing button	Button	Applies criteria and refreshes the grid.
Select	Existing - updated	Checkbox	Adds/removes a source record from the proposed BAR; disabled for a completed/generated row.
Generate BAR	Existing - updated	Button	Requires at least one eligible selection; opens the new confirmation control.
Print Batch Details	Existing button	Button	Requires a valid selection; opens the Batch Assembly Record details and print action.
Total BNS Batch	Existing indicator	Calculated display	Shows result/context B&S batch total.
Total Composite Batch	Existing indicator	Calculated display	Shows composite batch total.
Total Packs	Existing indicator	Calculated display	Shows pack total.
Not Printed BAR	Existing indicator/button	Status display	Identifies items without a generated BAR.

Grid column	Classification	Design behaviour
PRODUCT_STATUS	Existing grid column	Displays Completed after BAR generation; otherwise source status.

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Grid column	Classification	Design behaviour
SITE	Existing grid column	Existing source-record value displayed without edit.
COUNTRY	Existing grid column	Existing source-record value displayed without edit.
PART_NO	Existing grid column	Existing source-record value displayed without edit.
FOREIGN_NAME	Existing grid column	Existing source-record value displayed without edit.
ECMA	Existing grid column	Existing source-record value displayed without edit.
STRENGTH	Existing grid column	Existing source-record value displayed without edit.
PACK_SIZE	Existing grid column	Existing source-record value displayed without edit.
BATCH_NO	Existing grid column	Controlled B&S batch identifier and primary workflow key.
EXPIRY_DATE	Existing grid column	Existing source-record value displayed without edit.
QUANTITY	Existing grid column	Source quantity; summed when compatible rows are combined.
IMP	Existing grid column	Existing source-record value displayed without edit.
INVOICE_NO	Existing grid column	Existing source-record value displayed without edit.
DESCRIPTION	Existing grid column	Existing source-record value displayed without edit.
WAREHOUSE_LO	Existing grid column	Existing source-record value displayed without edit.
CONTRACT_SUPP	Existing grid column	Existing source-record value displayed without edit.
PL	Existing grid column	Existing source-record value displayed without edit.
PRODUCT_ID	Existing grid column	Existing source-record value displayed without edit.
MFG_ID	Existing grid column	Existing source-record value displayed without edit.

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Grid column	Classification	Design behaviour
ACTION_COUNT	Existing grid column	Existing source-record value displayed without edit.
OBJID	Existing grid column	Existing source-record value displayed without edit.
SPL_FLAG	Existing grid column	Retained flag; may indicate braille requirement.
CATEGORY	Existing grid column	Route driver: Reboxing or Relabelling.
STOCK	Existing grid column	Existing source-record value displayed without edit.
REGULATORY	Existing grid column	Existing source-record value displayed without edit.
WFIMPLEMENTED	Existing grid column	Existing source-record value displayed without edit.
LEGAL_CATEGORY	Existing grid column	Existing source-record value displayed without edit.

3.1.2 Selection, Combination and Confirmation

Step 1 - Add or remove selection. Selecting a row stores its stable source record identifier. Clearing the checkbox removes it and displays Batch removed from the BAR combination.

Step 2 - Compatibility validation. For multiple rows, PART_NO/product name, BATCH_NO and EXPIRY_DATE must match. A mismatch clears the attempted selection and displays Batches Cannot Be Combined / Selected batch details do not match.

Step 3 - Related-batch warning. Where configured, a warning table displays Order No and Quantity. Yes accepts the relationship; No cancels the combination without generating a BAR.

Step 4 - Generation confirmation. A single selection opens Generate BAR Confirmation. Multiple selections open Combined BAR Confirmation and show the selected row count, Batch_No and summed total quantity.

Step 5 - Confirmation outcome. Confirm creates one BAR and closes the dialog. Cancel/X closes the dialog and leaves the source rows available without a generated record.

Rule	System design
Minimum selection	At least one non-completed eligible source row.
Compatible combination	Same PART_NO/product name, BATCH_NO and EXPIRY_DATE.

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Rule	System design
Combined quantity	Sum of selected source quantities; applied to label/leaflet and applicable carton/braille required quantities.
Source traceability	Retain record ID, invoice, warehouse and quantity for every selected source record.
Duplicate prevention	A completed/generated row is disabled; only one active controlled BAR may exist for the batch.
Confirmation audit	Generated By and Generated At come from the authenticated session and system timestamp.

3.1.3 Generated BAR Design

Page	Title	Content design
1	BAR cover	Product Name, Foreign Name, Strength, Pack Size, ECMA, PL No., Units per pack, Country of origin, Product Introduced, Leaflet Date, Date Revised, B&S Batch Number, Expiry Date, variation information and generation attribution.
2	B&S Batch Add - Line Clearance	Three electronic line-clearance checks, comments, completed by and date/time.
3	Label Printing	Line completions, required/actual/extra print actions, user/date-time and details.
4	Leaflet Printing	Leaflet document/print and completion record.
5	Label Evidence Attachment	Test print, first production label, last production label and additional/extra evidence spaces.
6	Braille Printing	Braille label/declaration actions and completion.
7	Leaflet Folding	Folding quantity, status and attribution.
8	Carton Issuing	Required and extra carton issue records.
9	Pre-Assembly	Controlled downstream placeholder/audit presentation.
10	Production Control	Controlled downstream placeholder/audit presentation.
11	Assembly	Controlled downstream placeholder/audit presentation.
12	Post-Assembly QC	Controlled downstream placeholder/audit presentation.
13	BAR Audit Trail	Action, user, date/time and event details for the batch.

Reboxing attachment: When a completed Change of Pack Size record exists, it is appended after the controlled 13 BAR pages as an unnumbered attachment.

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View Generated BAR: Available after generation. Opens the populated BAR in view-only mode and displays the number of B&S checks captured.

Print Generated BAR: Available from the controlled document area. Opens the populated BAR with a print action and records user/date-time; printing does not change workflow completion.

Continuation pages: Label Evidence Attachment, BAR Continuation, Cold Chain Continuation and route-specific Change of Pack Size pages use the selected batch/product header and are audited when printed.

3.1.4 Line Clearance and Sign-off

Control	Design
PCL completeness/invoice check	Mandatory checkbox: PCL checked for completeness and invoice attached.
Batch/expiry check	Mandatory checkbox: batch number and expiry date are correct.
Product/pack/reference check	Mandatory checkbox: product name, strength, pack size and ECMA are correct.
Comments	Editable until sign-off; retained with the B&S line-clearance record.
BAR Created By	Read-only authenticated creator.
Created Date / Time	Read-only BAR generation timestamp.
View Generated BAR	Opens current controlled BAR; does not complete the stage.
User Sign Off	Disabled until all three checks are selected. On success records user/date-time, locks checks/comments, removes the batch from active B&S Batch Add and adds it to Label Printing.

3.2 Printing Module - Common Design

Label, Leaflet and Braille Printing use the same menu, queue, product/document context, unified Print pop-up, audit and completion principles. Carton Issuing shares menu, queue, product/document and audit presentation but records an issue rather than a printer run.

Control	Classification	Detailed design
Printer Menu	Existing - updated	Common entry screen for the four controlled options.
Label Printing	Existing tile/button	Opens Label Printing List.
Leaflet Printing	Existing tile/button	Opens Leaflet Printing List.
Carton Issuing	Existing tile/button	Opens Carton Issuing List.

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Control	Classification	Detailed design
Braille Printing	Existing tile/button	Opens Braille Printing List.
Active count	New calculated indicator	Count of untouched eligible records for the option.
In Progress count	New calculated indicator	Count with saved/printed/partially completed activity.
Back to Batch Queue	Existing common button	Returns to the active module list without completing or discarding saved records.

Queue control/column	Classification	Detailed design
B&S Batch Number	Existing field/column	Search input and primary displayed batch identifier.
MFG Lot No.	Existing field/column	Search input and manufacturing-lot display.
Search	Existing button	Applies both search inputs to the selected queue.
Product Name	Existing column	Approved product name.
Strength	Existing column	Product strength.
Pack Size	Existing column	Approved pack size.
ECMA	Existing column	Approved reference.
Expiry Date	Existing column	Batch expiry.
Required Qty	Existing data, module-specific	Label, leaflet, carton or braille required quantity according to queue.
Status	New calculated column	Active or In Progress in the module; the Label list may also show the downstream workflow position after completion.
Select queue row	Existing - updated action	Click/open the applicable detail screen; only eligible route-controlled records are listed.

Product information field	Classification	Source / behaviour
Product Name / Foreign Name	Existing common fields	Read-only approved product master data.
Strength / Country of origin	Existing common fields	Read-only product master data.

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Product information field	Classification	Source / behaviour
Pack Size / Units per pack	Existing common fields	Read-only packaging configuration.
B&S Batch Number / MFG Lot No.	Existing common fields	Read-only controlled batch identifiers.
ECMA / PL No.	Existing common fields	Read-only approved references.
Expiry Date / Quantity	Existing common fields	Read-only batch data.
Product Introduced	Existing common field	Read-only product introduction value/date.
Leaflet Date / Date Revised	Existing common fields	Read-only approved leaflet/version dates.

Document button	Classification	Detailed design
View Carton Artwork	Existing common button	Opens approved carton artwork in view-only preview.
View Peel Artwork	Existing common button	Opens approved peel/label artwork.
View Braille Artwork	Existing common button	Opens approved braille artwork/declaration.
View Mock-up	Existing common button	Opens approved product mock-up.
View BAR	Existing - updated access	Opens the controlled BAR for the selected batch.
Print Generated BAR	New common button	Opens populated BAR with audited print action.
Print Label Attachment	New common button	Prints additional label-evidence continuation page.
Print BAR Continuation	New common button	Prints general BAR continuation page.
Print Cold-chain Page	New common button	Prints temperature/location/checker continuation table.
Print Change of Pack Size	New route-specific button	Visible/available for Reboxing with the completed form.

3.2.1 Unified Print Pop-up - Entire Component Is New

The module-level Print button is an Existing replicated button. The complete pop-up opened by that button, including every field, selection and dialog action below, is New

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functionality. Existing/New is communicated by text only; all rows use the same font and size.

Control	Classification	Detailed design
Print pop-up	New pop-up	Opens only for Label, Leaflet or Braille and the selected component line.
Quantity Needed	New read-only field	Displays the approved required quantity for the selected line.
Quantity to Print	New entry field	Blank on open; numeric; minimum 1; must be a positive whole number.
Category	New controlled selection	Mandatory. Placeholder Select category; values Test Print, Actually Print and Extra Print.
Reason	New conditional selection	Disabled by default. Enabled and mandatory only for Extra Print. Default display Not required.
Reason values	New controlled values	Print damage; Line setup waste; Reconciliation correction; Printing alignment check; Supervisor approved extra.
Print (dialog)	New controlled button	Disabled until the required inputs are valid; executes the selected category behaviour.
Close / X	New dialog action	Closes the pop-up without executing or changing the line.
Preview area	New preview presentation	Label/Braille show component, product, strength, pack size, batch and expiry; Leaflet opens the controlled leaflet PDF after submission.

Design area	Condition	System behaviour
Validation	No category	Message: Select a print category before printing. No record is created.
Validation	Quantity not a positive integer	Message: Enter a whole print quantity greater than zero.
Validation	Extra Print without reason	Message: Select a reason before printing extra copies.
Test Print	Label/Braille	Record test user/date/time/quantity. Do not reduce required quantity and do not add a normal audit event.
Test Print	Leaflet	Record test details and open the test-category leaflet PDF; Test Print is excluded from the BAR audit table.

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Design area	Condition	System behaviour
Actually Print	Label/Braille	Apply to remaining required quantity. Quantity cannot exceed the remaining required quantity. Record print run and update remaining amount.
Actually Print	Leaflet	Record actual-print item, PDF-open state and print-run details; this enables Mark as Done.
Extra Print	All shared print modules	Accumulate extra quantity separately, require a controlled reason and create an attributable audit event.
Post-completion print	All shared print modules	Print categories remain available, but subsequent prints do not replace or alter the completed Batch Record Summary.
Line Completion	Label/Braille	Mark as Done enables when remaining required quantity is zero.
Line Completion	Leaflet	Mark as Done enables only after an Actually Print event.
Print Done	Module	Enabled only when every mandatory displayed component line is Done; records final user/date-time and routes the batch.

3.2.2 Common Print Data and Audit Design

Data element	Design
Batch number / module / item	Identifies the controlled record and component line.
Quantity needed	Required quantity at the time of the action.
Quantity to print	Positive whole quantity submitted in the dialog.
Category	Test Print, Actually Print or Extra Print.
Reason	Not required except controlled value for Extra Print.
Print totals / remaining	Required actual quantities are accumulated by item; remaining quantity is derived and never negative.
Test-print map	Stores item, quantity, user and date/time separately from required production totals.
Extra-print runs	Append-only list by item with quantity, reason, user and date/time.
Item completion	Status, completing user and date/time for each line.
Finalisation	Finalized flag, finalized by and finalized at; does not remove prior run details.

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Data element	Design
Batch summary entries	Non-test actions occurring before line completion are presented on the controlled BAR summary.
Printer audit trail	Action type, batch, user, date/time and human-readable details.

3.3 Label Printing Module

Design item	Detailed design
Queue eligibility	Generated BAR exists and B&S User Sign Off is complete; completed records remain available only according to downstream/status presentation.
Reboxing rows	Carton Label; End of Pack Label; Security Seals.
Relabelling rows	Blister / Pack Label; Obscure Label.
Row columns	Label, Reference, Size, Quantity, Location, In Hand Quantity, Print, Line Completion, Completed By / Date.
Reference/size/quantity	Derived from approved product/route data. Security Seals use the controlled seal reference and quantity.
Print button	Existing replicated button opening the new common Print pop-up.
Line completion	Mark as Done is disabled while required quantity remains; Done locks the normal completion and displays attribution.
Print Done	Enabled after every displayed route-derived row is Done.
Output routing	Leaflet-required batches move to Leaflet Printing; otherwise the next configured stage.

3.3.1 Reboxing Change of Pack Size Gate

Control / section	Detailed design
Change of Pack Size form	Existing form represented electronically for Reboxing only.
Received As	Records received packaging configuration and electronic sign-off.
Assembled As	Records target configuration and electronic sign-off.
Print gate	All label Print buttons remain disabled and the gate message is displayed until both sections are complete and signed.
Locking	Once label printing has started, the supporting form is treated as the controlled source and cannot be silently altered.

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Control / section	Detailed design
BAR attachment	Completed form is appended after BAR page 13 as an unnumbered route-specific attachment.

3.4 Leaflet Printing Module

Design item	Detailed design
Queue eligibility	Label Printing completed and the product route requires a leaflet.
Table columns	Leaflet, Reference, Size, Quantity, Location, Print, Line Completion, Completed By / Date.
Print action	Opens the common pop-up. After submission, the controlled leaflet PDF opens with category, quantity and reason.
PDF audit	The approved PDF opening is attributable. Non-test print actions are added to BAR/audit history before completion.
Test/master control	Required test/master review is completed before controlled completion; rejected evidence remains traceable as exception evidence.
Mark as Done	Disabled until an Actually Print event exists for the line.
Extra Print	Positive whole quantity and controlled reason; accumulated separately.
Print Done	Requires all leaflet rows Done.
Routing - Reboxing	Move to Carton Issuing.
Routing - Relabelling with braille	Move to Braille Printing.
Routing - other leaflet route	Move to Leaflet Folding or configured next stage.

3.5 Carton Issuing Module

Control / column	Classification	Detailed design
Carton Issuing List	Existing - updated eligibility	Only Reboxing batches that completed Leaflet Printing.
Carton Reference	Existing field	ECMA/part number or approved carton reference.
Required Qty	Existing calculated field	Required carton quantity for the batch.
On Hand Quantity	Existing field	Available carton stock.

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Control / column	Classification	Detailed design
Issue Extra	Existing - updated button	Disabled until required issue is confirmed. Opens extra issue dialog.
Extra quantity	Controlled entry	Positive whole number.
Extra reason	Controlled selection	Mandatory for extra issue; values include Carton damage and other configured issue reasons.
Location Number	Existing field	Warehouse/location used for the issue.
Confirmed By	New audit field	Pending until Done; then authenticated user and date/time.
Done	Existing - updated button	Confirms the normal required issue once and becomes disabled.
Completion count	New indicator	0/1 before confirmation; 1/1 after confirmation.
Output	BAR/audit update and Carton prerequisite satisfied for Leaflet Folding.	

3.6 Braille Printing Module

Design item	Detailed design
Queue eligibility	Relabelling, Leaflet Printing completed and braille-required flag is true.
Mandatory rows	Braille Label and Braille Declaration Copy where configured.
Braille Label source	Reference ECMA; size pack size; quantity approved braille quantity.
Declaration Copy source	Reference PL No.; size Master copy; quantity 1.
Table columns	Braille Label, Reference, Size, Quantity, Location, In Hand Quantity, Print, Line Completion, Completed By / Date.
Print behaviour	Common Test/Actually/Extra Print pop-up and validation.
Mark as Done	Enabled when remaining required quantity for the row is zero.
Print Done	Requires every displayed braille/declaration row Done.
Output	BAR/audit update and Braille prerequisite satisfied for Leaflet Folding.

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3.7 Leaflet Folding Module

Queue control/column	Classification	Detailed design
B&S Batch Number	Existing search/column	Searches and displays the controlled batch.
MFG Lot No.	Existing search/column	Searches and displays the manufacturing lot.
Search	Existing button	Applies both queue criteria.
Product Name	Existing column	Read-only product name.
Strength	Existing column	Read-only strength.
Pack Size	Existing column	Read-only pack size.
ECMA	Existing column	Read-only approved reference.
Expiry Date	Existing column	Read-only batch expiry.
Required Qty	Existing column	Required leaflet/folding quantity.
Status	New calculated column	Active or Completed/current controlled state.
Select row	Existing - updated action	Opens detail and preserves the selected batch context.

Detail control/column	Classification	Detailed design
Product Information/Documents	Existing common display - updated access	Uses the common product and controlled-document presentation.
Leaflet Reference	Existing field	Approved ECMA/leaflet reference.
Leaflet Size	Existing field	Approved pack-size leaflet format.
Required Qty	Existing field	Quantity that must be folded.
On Hand Quantity	Existing field	Available leaflet quantity.
Folded By	New audit field	Pending confirmation then authenticated user.
Date / Time	New audit field	Pending confirmation then system timestamp.
Done	Existing - updated button	Single-use normal completion. Records quantity/user/date-time, updates BAR/audit, removes batch from active queue and routes to Pre-Assembly.

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Detail control/column	Classification	Detailed design
Back to Batch Queue	Existing button	Returns without completing.

3.7.1 Eligibility Correction Required by Approved FS

The wireframe list currently derives candidates from Leaflet Printing completion. The implemented design shall add route-specific prerequisite checks before a record is returned to the Leaflet Folding queue: Reboxing requires completed Carton Issuing; Relabelling with braille required requires completed Braille Printing; relabelling without braille requires completed Leaflet Printing only. This correction prevents premature folding.

Validation	Failure behaviour
Batch identity not valid	Do not complete; retain Active status and selected batch context.
Leaflet reference or size not approved/mismatched	Do not complete; display validation and retain the record in the queue.
Required folded quantity missing/invalid	Do not complete or create attribution.
Route prerequisite incomplete	Do not list the batch in the Leaflet Folding queue.
Already completed	Done remains disabled; retained history is view-only through authorised access.

3.8 Workflow Status and Routing Design

Status	Module	Meaning	Next action
Not Printed BAR	B&S Batch Add	Eligible source exists; BAR not generated.	Generate BAR
Generated	B&S Batch Add	Controlled BAR exists; line clearance pending.	Complete checks
B&S Line Clearance Complete	B&S Batch Add	All checks signed.	Automatic handoff to Label Printing
Active	Printing/Folding queue	Eligible; no saved action.	Open batch
In Progress	Printing queue	At least one saved/test/actual/line action.	Continue remaining lines
Label Printing complete	Label Printing	All route-derived label lines Done and Print Done recorded.	Route by leaflet requirement

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Status	Module	Meaning	Next action
Leaflet Printing complete	Leaflet Printing	All leaflet lines Done and Print Done recorded.	Carton, Braille or Folding branch
Carton Issuing complete	Carton Issuing	Normal carton issue confirmed.	Satisfy Reboxing folding prerequisite
Braille Printing complete	Braille Printing	All braille rows Done and Print Done recorded.	Satisfy applicable folding prerequisite
Leaflet Folding complete	Leaflet Folding	Done recorded with quantity/user/date-time.	Handoff to Pre-Assembly

3.9 Audit, Data Integrity and Record Design

Audit event	Minimum retained content
BAR generation	Batch, source record IDs, total quantity, generated by/date-time, BAR version/status.
B&S line clearance	Three check results, comments, completed by/date-time, signed-off flag.
Document view/print	Document title/version/reference, action, batch, user and date/time.
Print run	Module, component, quantity needed, quantity selected, category, reason, user and date/time.
Line completion	Component, status Done, user and date/time.
Print Done	Module finalisation, batch, final user/date-time and retained line/run data.
Carton issue	Required and extra quantities, reason, location, user/date-time and completion.
Leaflet Folding	Reference, size, required/folded quantity, user/date-time and completed status.
Routing	Previous and new stage/status with batch and timestamp.
Correction/reprint	Append new event and reason; never overwrite the original controlled action.

Persistence: Records shall use the existing approved PLPI database/file controls. Wireframe local-storage objects are demonstrators of required state only and are not the production persistence design.

Concurrency: A controlled batch/stage shall be locked against conflicting completion. A stale screen must revalidate status before committing.

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Read-only completion: Completed normal actions are disabled. Additional print or correction uses a separate authorised, reasoned, append-only action.

Time: Server-controlled date/time shall be used in production; client display uses the agreed UK-readable format.

4. Additional Non-Functional Requirements

4.1 Audit Trail

Retain all controlled search-selection outcomes where relevant, generation, view, print, completion, routing, error/retry and correction events with batch/item/user/role/date-time and reason/details.

4.2 Availability

Functions are available to authorised operational and support roles during agreed hours.

4.3 Capacity Limits

Support expected eligible batch volumes, controlled BARs, queue records, component lines, print runs, extra reasons, folding records and retained history.

4.4 Performance

Search, BAR view/generation, document opening, print submission, line completion and queue handoff operate within agreed operational response times.

4.5 Recoverability

Saved progress remains available without duplication. Failed print/PDF/routing actions do not falsely advance status.

4.6 Security

Existing role-based authentication. Completed records are read-only except through authorised correction/reprint processes.

4.7 Error Handling

Failed validation blocks the action, shows a clear message and leaves the batch at the current controlled stage.

4.8 Usability

Consistent queues, product/document panels, print pop-up and completion language; no overlap or truncation.

4.9 Accuracy and Validity

Source selections, identifiers, quantities, references, reasons, checks and route prerequisites remain linked to the committed batch record.

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4.10 Access and Responsibilities

Operational actions are restricted to the stage-owning role; QA/RP/Operations/IT access follows approved responsibility.

4.11 Testing

Cover eligibility, combination, duplicate prevention, 13-page BAR, line-clearance gate, all popup categories/validations, module rows, route branches, audit and folding prerequisites.

4.12 Controlled Documents and Training

SOPs/work instructions describe BAR creation, common print controls, category/reason use, line/Print Done, carton issue and folding completion.

4.13 Support and Administration

IT supports access, configuration, document/artwork links, printer connectivity, queues, audit retrieval, backup/recovery and incidents.

5. URS / FS / DS Traceability

URS ID	FS reference	DS reference	Feature
4.1.2	FS 3.1	DS 3.1.1	B&S search and selection
4.1.3	FS 3.1	DS 3.1.1	Selected batch/source display
4.1.4	FS 3.1	DS 3.1.2	Compatible source combination
4.1.5	FS 3.1	DS 3.1.2-3.1.3	Controlled BAR creation
4.1.6	FS 3.1	DS 3.1.3-3.1.4	BAR verification and line clearance
4.1.7	FS 3.1	DS 3.1.4	B&S sign-off and Label handoff
4.2.1	FS 3.2	DS 3.2	Printer options/menu
4.2.2	FS 3.2	DS 3.2	Printing queues and product display
4.2.3	FS 3.2	DS 3.2	Controlled BAR/artwork/documents
4.2.4	FS 3.3	DS 3.3	Route-derived label lines
4.2.5	FS 3.2-3.4, 3.6	DS 3.2.1-3.2.2, 3.3-3.4, 3.6	Common printing and popup
4.2.6	FS 3.4-3.6	DS 3.4-3.6	Route branching to carton/braille
4.2.7	FS 3.5-3.6	DS 3.5-3.6	Carton and braille controls
4.2.8	FS 3.2-3.7	DS 3.2-3.8	Completion and folding handoff
4.3.1	FS 3.7	DS 3.7.1	Folding eligibility

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URS ID	FS reference	DS reference	Feature
4.3.2	FS 3.7	DS 3.7	Folding search and display
4.3.3	FS 3.7	DS 3.7	Folding document access
4.3.4	FS 3.7	DS 3.7	Batch/leaflet confirmation
4.3.5	FS 3.7	DS 3.7	Folded quantity and completion gate
4.3.6	FS 3.7	DS 3.7, 3.9	Folding attribution/history

6. Design Verification and Acceptance Considerations

Verification area	Acceptance consideration
B&S grid	All filters, 28 source columns, totals and both action buttons render within the screen and retain Existing classification.
Combination	Matching records combine and sum quantity; mismatch, no-selection and cancel paths do not create a BAR.
BAR	One controlled 13-page record is populated; Reboxing attachment is unnumbered; view/print and audit work as designed.
Line clearance	Three checks gate User Sign Off; sign-off locks the record and routes to Label Printing.
Printer common	Four menu options, counts, shared queue, product panel, document buttons and Back action are consistent.
Print pop-up	The entire pop-up is New; all fields/buttons use common font size; category/quantity/reason validation and post-completion behaviour are correct.
Label	Correct rows are derived for Reboxing/Relabelling; Change of Pack Size gate works; line completion and Print Done are controlled.
Leaflet	Controlled PDF opens; actual print enables Mark as Done; route branches correctly.
Carton	Required issue is single-use; Issue Extra is blocked until confirmation and requires quantity/reason.
Braille	Only eligible batches appear; both configured rows complete before Print Done.
Folding	Route prerequisites are enforced; Done records quantity/user/date-time and hands off once.
Audit/security	Every controlled event is attributable; completed records cannot be overwritten; role and concurrency checks are enforced.